

Experiment - Study of LAN Topology, Network devices and Address configuration.

Objective :-

1. To study the LAN topology used in college campus.
2. To identify networking device & understand their function.
3. To analyze wired & wireless communication method used in the network.
4. To measure speed of Network for wired & wireless connections.
5. To obtain MAC, IP Address & subnet mask.

Problem Statement ↓

Draw & identify LAN topology of college campus & clearly label all components such as router, switches etc.

The college campus network observed this study consist of :

- Layer 3 core switches in the server room.
- Managed by POE + switches.
- Wireless Access point (APs)
- Ethernet port in classroom & laboratories.

Network Topology-

Hybrid Hierarchical topology which combines:

- (1) Tree Topology
- (2) Extended Star Topology
- (3) Ring Topology (server Room)

The network hierarchical layer.

②

- 1 Core layer
- 2 Distribution layer.
- 3 Access layer

→ Core layer

Core layer contains server room and server room have firewall, Router, 2-L3t switches, patch Rack, server, inverter, etc

→ Distribution layer

- Distribution layer contains 4-switches.

- The 4-switches are

- (1) Main Build → R7 (Room no.-213, 2nd Floor)
- (2) Main Build → R1 (Room no.-301, 3rd Floor)
- (3) Main Build → R6 (Room no.-113, 1st Floor).
- (4) Main Build → R2 (Room no.-201, 2nd Floor).

- This 4-switches contains —

- (1) 18 port Gigabit Easy Smart Switch with 16 port . TL-SG1218MPE

- (2) SG3428XMP → 24 port Gigabit & 4 port 10 GE SFP+, L2 + Managed switches with 24 port POE+.

→ Now let's discuss these 2 above components —

i. TP-Link TL-SG1218MPE

* likely used for : CCTV camera, wifi access point etc.

- * 18 port Gigabit "Easy Smart" switch.
- * 16 port support POE+.

② TP-Link Omada SG3428XMP

- * Higher-end managed layer 2+ switch
- * 24 POE+ port, uplinks, 10G SFP
- * Usually acts as a main distribution for each floor.

Access Layer

Access layer contains 3-switches i.e

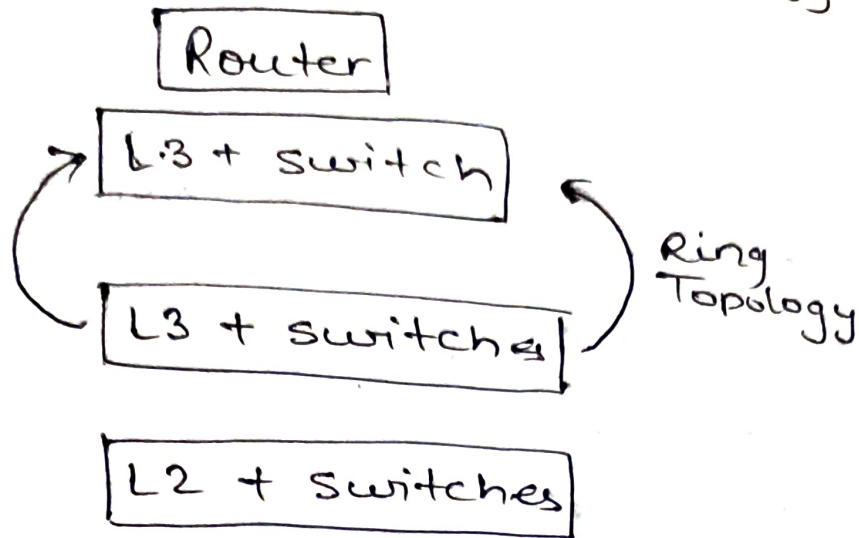
- (1) Main Build R1 - [Room no. 301, 3rd floor]
- (2) Main Build R4 - [Room no. -05, Ground floor]
- (3) Main Build R5 - [Room no. -18, Ground floor]

- * Every switches act as access layer which helps to provide network connectivity to all classroom, lab, faculty room etc.

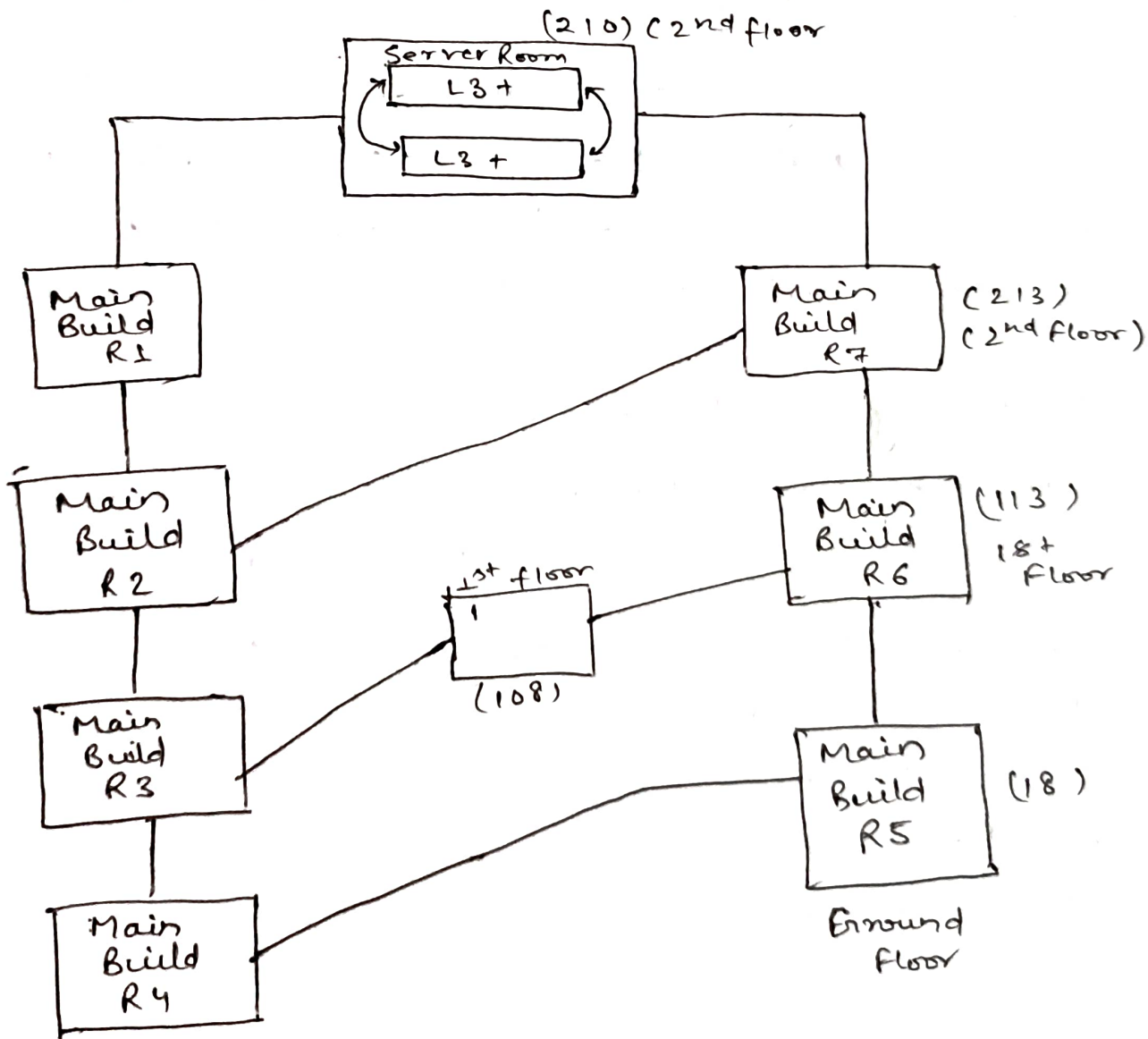
Access layer directly connect:

1. Ethernet Port
2. Wireless AP
3. Lab System
4. Outdoor Wireless AP (ex-018)

SERVER Room Architecture / Topology

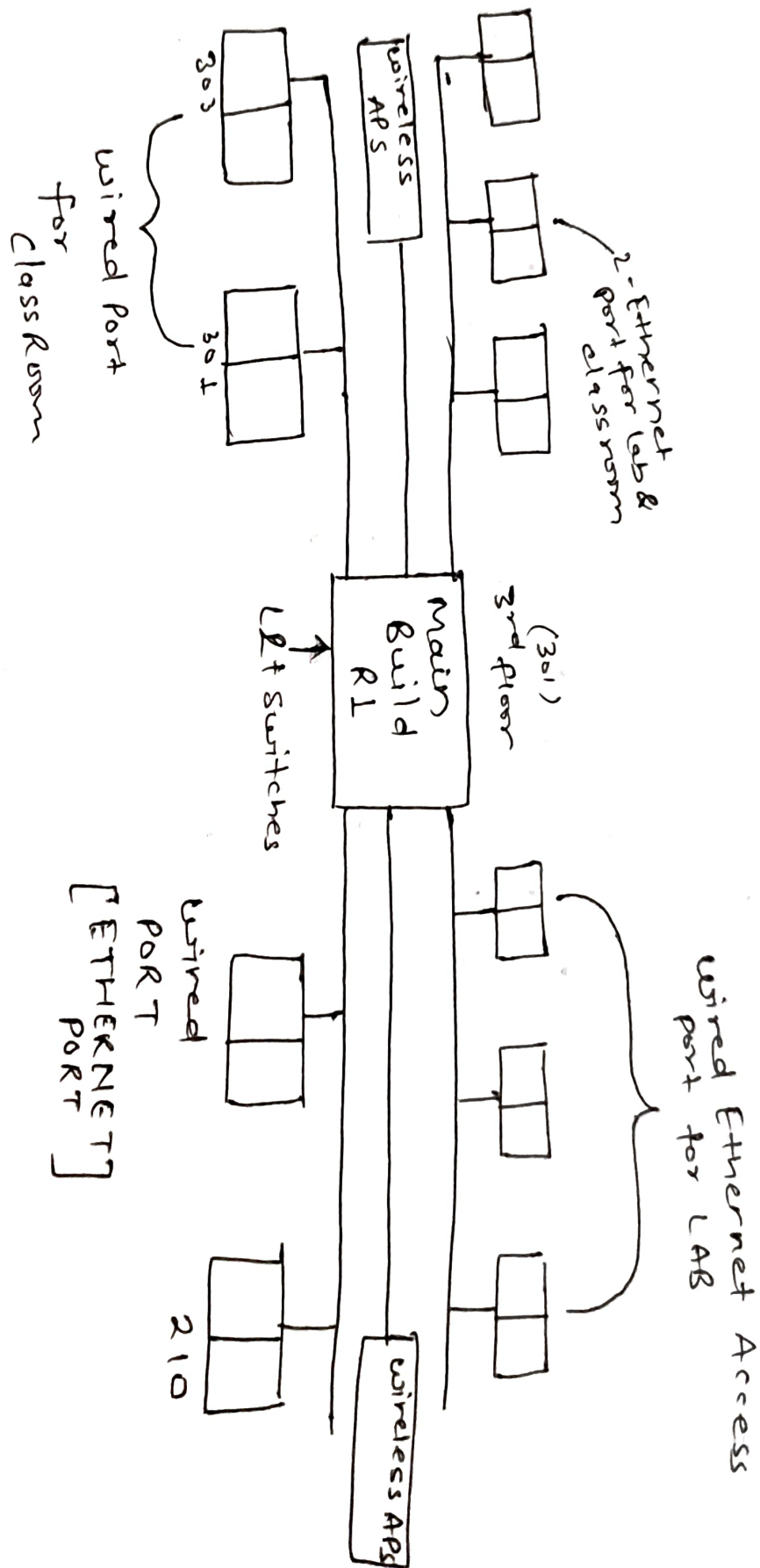


Topology-



- R1, R7, R2, R6 → Distribution layer
- Server Room → Core layer
- R1, R2, R3, R4, R5, R6, R2 → Access layer

Access Layer Switch (3rd floor 301)



```
Command Prompt
C:\Users\NANDKISHOR GUPTA>netsh wlan show interfaces

There is 1 interface on the system:

Name                : Wi-Fi
Description         : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter
GUID               : f160490e-1787-4c4d-a0da-adbec301d5bd
Physical address    : 38:8d:3d:1c:97:1a
Interface type      : Primary
State               : connected
SSID               : GECK-Campus-WiFi
AP BSSID           : b0:19:21:30:b8:a6
Band                : 2.4 GHz
Channel             : 11
Network type        : Infrastructure
Radio type          : 802.11ax
Authentication      : WPA3-Personal
Cipher              : CCMP
Connection mode     : Auto Connect
Receive rate (Mbps) : 143
Transmit rate (Mbps) : 86
Signal              : 90%
Profile             : GECK-Campus-WiFi
QoS MSCS Configured : 0
QoS Map Configured  : 0
QoS Map Allowed by Policy : 0

C:\Users\NANDKISHOR GUPTA>
```

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```
Command Prompt
Network type      : Infrastructure
Radio type       : 802.11ax
Authentication    : WPA3-Personal
Cipher           : CCMP
Connection mode   : Auto Connect
Receive rate (Mbps) : 143
Transmit rate (Mbps) : 86
Signal           : 90%
Profile          : GECK-Campus-WiFi
QoS MSCS Configured : 0
QoS Map Configured : 0
QoS Map Allowed by Policy : 0

C:\Users\NANDKISHOR GUPTA>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2409:40e4:11e5:130e:1e55:ec1:7ce9:ce6c
Temporary IPv6 Address. . . . . : 2409:40e4:11e5:130e:7d8c:1305:f76e:e689
Link-local IPv6 Address . . . . . : fe80::5012:b7c6:76ff:c74c%16
IPv4 Address. . . . . : 10.99.254.207
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::20a4:56ff:fe32:4b18%16
                             10.99.254.201

C:\Users\NANDKISHOR GUPTA>|
```

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```
Command Prompt
IPv6 Address. . . . . : 2409:40e4:11e5:130e:1e55:ec1:7ce9:ce6c
Temporary IPv6 Address. . . . . : 2409:40e4:11e5:130e:7d8c:1305:f76e:e689
Link-local IPv6 Address . . . . . : fe80::5012:b7c6:76ff:c74c%16
IPv4 Address. . . . . : 10.99.254.207
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::20a4:56ff:fe32:4b18%16
10.99.254.201

C:\Users\NANDKISHOR GUPTA>getmac
INFO: No network adapters found.

C:\Users\NANDKISHOR GUPTA>getmac
INFO: No network adapters found.

C:\Users\NANDKISHOR GUPTA>getmac
INFO: No network adapters found.

C:\Users\NANDKISHOR GUPTA>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::5012:b7c6:76ff:c74c%16
    IPv4 Address. . . . . : 10.85.19.25
    Subnet Mask . . . . . : 255.255.248.0
    Default Gateway . . . . . : 10.85.16.1

C:\Users\NANDKISHOR GUPTA>
```

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Command Prompt
C:\Users\NANDKISHOR GUPTA>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-8CIPEH56
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter #4
Physical Address. . . . . : 3E-8D-3D-1C-97-1A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter #5
Physical Address. . . . . : 3A-8D-3D-1C-97-1A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 38-8D-3D-1C-97-1A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::5012:b7c6:76ff:c74c%16(Preferred)
IPv4 Address. . . . . : 10.85.19.25(Preferred)
Subnet Mask . . . . . : 255.255.248.0
Lease Obtained. . . . . : 29 May 2026 17:57:43
Lease Expires . . . . . : 30 May 2026 06:06:17

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```
Command Prompt
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Wireless LAN adapter Wi-Fi 4:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter #4
Physical Address. . . . . : 3E-8D-3D-1C-97-1A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 5:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter #5
Physical Address. . . . . : 3A-8D-3D-1C-97-1A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 38-8D-3D-1C-97-1A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::5012:b7c6:76ff:c74c%16(Preferred)
IPv4 Address. . . . . : 10.85.19.25(Preferred)
Subnet Mask . . . . . : 255.255.248.0
Lease Obtained. . . . . : 29 May 2026 17:57:43
Lease Expires . . . . . : 30 May 2026 06:06:17
Default Gateway . . . . . : 10.85.16.1
DHCP Server . . . . . : 10.85.16.1
DHCPv6 IAID . . . . . : 305696061
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-AB-18-04-38-8D-3D-1C-97-1A
DNS Servers . . . . . : 8.8.8.8
                        1.1.1.1
```

Command Prompt

Primary Dns Suffix
No
IP
WI

Task Manager

Performance

CPU
3% 1.21 GHz

Memory
7.1/15.7 GB (45%)

Disk 0 (C: D: E:)
SSD (NVMe)
9%

Wi-Fi
Wi-Fi
S: 32.0 R: 64.0 Kbps

GPU 0
Intel(R) Iris(R) Xe Gra...
0%

Wi-Fi

Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter

Throughput

500 Kbps

450 Kbps

60 seconds

Send

32.0 Kbps

Receive

64.0 Kbps

Adapter name:

Wi-Fi

SSID:

GECK-Campus-WiFi

Connection type:

802.11ax

IPv4 address:

169.254.121.24

IPv6 address:

fe80:5012:b7c6:76ffc74c%16

Signal strength:

Subnet Mask : 255.255.248.0

Lease Obtained. : 29 May 2026 17:57:43

Lease Expires : 30 May 2026 06:06:17

Default Gateway : 10.85.16.1

DHCP Server : 10.85.16.1

DHCPv6 IAID : 305696061

DHCPv6 Client DUID. : 00-01-00-01-2F-AB-18-04-38-8D-3D-1C-97-1A

DNS Servers : 8.8.8.8

1.1.1.1

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Q. Role of Subnet mask in network?

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

It is important for :-

- (1) Network Segment
- (2) Efficient Routing
- (3) Communication b/w devices

OBSERVATION

During the study of the college network infrastructure :-

- (1) A centralized server room was used as network backbone.
- (2) Two L3 switches were connected in redundant cyclic structure.
- (3) Managed TP-link POE + switches were used in network cabinets.

→ Advantage of Existing Network.

- Scalable architecture
- Centralized management
- Supports PoE+ device
- Easy expansion of floors & Classroom
- Support both wireless & wired.

Conclusion —

→ The observed campus LAN follow a hybrid hierarchical topology consist of tree, extended star, Ring. The network uses managed PoE switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.